This document contains syllabi of my taught courses and, whenever available, student evaluations. Specifically, the causes taught be me are

- **Statistics 1 (part 1),** University of Zurich, undergraduate (in German)
- **Dirty Data: Outliers and Missing Values in Statistics and Psychology**, University of Zurich, graduate
- **Academic skills**, Erasmus University, undergraduate
- **Seminar in machine learning**, Erasmus University, undergraduate
- **Machine learning**, Erasmus University, graduate
- **Seminar in econometrics**, Erasmus University, undergraduate
- I**ntroduction to multivariate statistics**, Erasmus University, undergraduate

Descriptions of my roles are provided in my CV. Furthermore, I supervised a total of five master theses and 16 bachelor theses as primary supervisor. Details are provided in my CV.

Course Catalog Fall Semester 2026

Printing Date: Nov 4, 2026

Link: https://studentservices.uzh.ch/uzh/launchpad?sap-language=EN&sap-ui-language=EN#ZVVZ-display&/details/2026/003/E/51325974/50000007/Philosophische%2520Fakult%25C3%25A4t/50890577/Bachelor%2520of%2520Science%2520in%2520Psychologie%2520(RVO19)/50893138/Psychologie///50942312/Einf%25C3%25BChrung%2520Statistik%252C%2520Emotions-%252C%2520Motivations-%252C%2520Sozialpsychologie

Event Statistik 1 (Teil 1)
06VL200-001a / Course / 2,817

Description:

Business event contents:
In dieser Veranstaltung werden Grundlagen, Voraussetzungen, praktische Anwendung und Interpretation der wichtigsten deskriptiven und einführenden inferenzstatistischen Verfahren behandelt, wie z.B. die grafische Darstellung von Verteilungen, statistische Kennwerte, Schätzung, Konfidenzintervalle und Hypothesentests. Die praktische Anwendung und Interpretation der Verfahren wird in begleitenden Tutoraten vertieft und eingeübt.

Notes:
Es wird ein Podcast aufgezeichnet, Zugang über OLAT. Hinweis: Es ist möglich, dass einzelne Vorlesungen nicht oder nicht störungsfrei aufgezeichnet werden. Studierende verzichten deshalb auf eigenes Risiko auf den Besuch einer Veranstaltung oder auf das Erstellen eigener Notizen. Bei inhaltlichen Widersprüchen haben die als prüfungsrelevant deklarierten Materialen (z.B. Folien) Vorrang vor den Podcasts. Bei Unklarheiten kontaktieren Sie bitte umgehend die Dozentin/den Dozenten. Die Aufnahmen dürfen nur für den Privatgebrauch verwendet werden. (G06)

Course Materials:	Podcast, Folien und weitere Unterrichtsmaterialien werden in einem OLAT- Kurs bereitgestellt. Ablauf und Unterrichtsmaterialien werden in der ersten Vorlesung besprochen.
Instructor:	Max Welz
Languages:	German
Open to Auditors:	Yes

Time/Room:

Date	Time	Room	Instructors
Sep 14, 2026	14:00 - 15:45	HAH-E-03 HAH-E-11	Max Welz
Sep 21, 2026	14:00 - 15:45	HAH-E-03 HAH-E-11	Max Welz
Sep 28, 2026	14:00 - 15:45	HAH-E-03 HAH-E-11	Max Welz
Oct 5, 2026	14:00 - 15:45	HAH-E-03 HAH-E-11	Max Welz
Oct 12, 2026	14:00 - 15:45	HAH-E-03 HAH-E-11	Max Welz
Oct 19, 2026	14:00 - 15:45	HAH-E-03 HAH-E-11	Max Welz
Oct 26, 2026	14:00 - 15:45	HAH-E-03 HAH-E-11	Max Welz

Nov 2, 2026	14:00 - 15:45	HAH-E-03 HAH-E-11	Max Welz
Nov 9, 2026	14:00 - 15:45	HAH-E-03 HAH-E-11	Max Welz
Nov 16, 2026	14:00 - 15:45	HAH-E-03 HAH-E-11	Max Welz
Nov 23, 2026	14:00 - 15:45	HAH-E-03 HAH-E-11	Max Welz
Nov 30, 2026	14:00 - 15:45	HAH-E-03 HAH-E-11	Max Welz
Dec 7, 2026	14:00 - 15:45	HAH-E-03 HAH-E-11	Max Welz
Dec 14, 2026	14:00 - 15:45	HAH-E-03 HAH-E-11	Max Welz

Component of:

Module: Introduction to Statistics and Emotional, Motivational and Social Psychology

Course Catalog Spring Semester 2026

Printing date: Nov 4, 2026

Link: https://studentservices.uzh.ch/uzh/launchpad?sap-language=EN&sap-ui-language=EN#ZVVZ-display&/details/2025/004/SM/51305737/50000007/Faculty%2520of%2520Arts%2520and%2520Social%2520Sciences/50896641/Master%2520of%2520Science%2520in%2520Psychology%2520(RVO19)/50905819/Psychology

Module Dirty Data: Outliers and Missing Values in Statistics and Psychology
06SM200o800b / Module

Description:

General description:
This seminar focuses on two practical issues that are commonly encountered in the statistical analysis of real-world psychological data, namely outliers and missing values. If addressed improperly, such "messy" data can invalidate the findings of any empirical study. The course's goal is to provide students with a toolbox of methods that they can use in their master theses and beyond to properly address outliers and missing values. The course adopts a "hands-on" approach in the sense that the students will learn how to deal with these issues from a practical perspective through computer exercises on real-world data, using the statistical programming environment R.
If time permits, the following topics will be covered:

1. Mechanisms and methods for missing values,
2. Outlier-robust methods for mean, variance, and correlation,
3. Outlier-robust linear regression,
4. Robustness in psychometrics

ECTS: 4

Notes:
Dieses Modul steht nur Studierenden des Master of Science in Psychologie (RVO19) offen. This module is only open to students of Master of Science in Psychology (RVO19).

Learning Outcome:
At the end of the course, students are able to analyze datasets containing missing values and/or outliers using the methods covered.
In particular, they can use software for outliers and missing values in the statistical programming environment R and know the "do"s and "don't"s of working with messy data. They are able to identify possible issues stemming from missing values and outliers, know how to overcome them methodologically, intuitively describe the covered methods, and know how apply them in practice by using software in R.

Languages of Instruction:
Englisch

Merkmale:	Target Group: Suitable for English Speakers
Responsible instructor:	Carolin Strobl

Requirements:

Assessment:

This seminar focuses on two practical issues that are commonly encountered in the statistical analysis of real-world psychological data, namely outliers and missing values. If addressed improperly, such "messy" data can invalidate the findings of any empirical study. The course's goal is to provide students with a toolbox of methods that they can use in their master theses and beyond to properly address outliers and missing values. The course adopts a "hands-on" approach in the sense that the students will learn how to deal with these issues from a practical perspective through computer exercises on real-world data, using the statistical programming environment R.
If time permits, the following topics will be covered:

1. Mechanisms and methods for missing values,

2. Outlier-robust methods for mean, variance, and correlation,
3. Outlier-robust linear regression,
4. Robustness in psychometrics

Portfolio

Individual takehome assignment ("Hausarbeit" in German) towards the end of the course which the students have one week to complete (100%). The students need their own laptop with up-to-date versions of R and R-Studio installed for the final assignment.

Grading Scale:	1-6, Half Grades Given
Repeatability:	Can't be repeated

Organisation:

Offer pattern:	1 Semester (One-Time)
Request period:	from Mi 14.01.2026 10:00 to Di 27.01.2026 24:00
Place allocation at the latest:	Sa 31.01.2026 22:00
Waiting list progression period:	from Sa 31.01.2026 22:00 to Do 12.02.2026 24:00
Cancellation period:	from Sa 31.01.2026 22:00 to So 15.02.2026 24:00
Organisation:	Department of Psychology
Offered in:	Spring Semester 2026

Component of:

Faculty of Arts and Social Sciences		
Psychology		
Single Major 120	Master of Science in Psychology (RVO19)	Advanced Research Methods

Courses:

Course	Begin/End	Time	Instructors
Seminar			
Dirty Data: Outliers and Missing Values in Statistics and Psychology 06SE200o800b	Feb 17, 2026 May 26, 2026	Tue 12:15-13:45	Max Welz

University of Zurich[UZH]

Educational Development

University of Zurich
Educational Development
Quality Management Education & Teaching
Hirschengraben 84
CH-8001 Zurich
www.le.uzh.ch/en

Private and Confidential

University of Zurich
LVE - Lehrveranstaltungsevaluation/Course Evaluation
To the instructors of the course:
"Dirty Data: Outliers and Missing Values in Statistics and Psychology"

Zurich in spring 2025

Evaluations of Teaching: Evaluation Report, Spring Semester 2025

Dear instructor(s)

Hereby you will find your evaluation report on results of your course:

„Dirty Data: Outliers and Missing Values in Statistics and Psychology“

The report contains frequencies of particular satisfaction aspects. Thus, features which need improvement from the students' perspective become recognizable. The satisfaction items with a scale show how satisfied the students are with the course overall. Furthermore, there are student comments included, if available.

Please feel free to contact us if you have any questions.

Sincerely
Quality Management Education & Teaching

These results are sent to the following instructor(s):

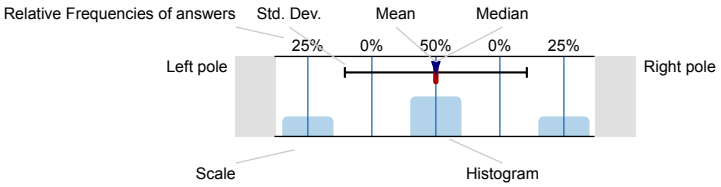
Max Felix Orson Welz

Dirty Data: Outliers and Missing Values in Statistics and Psychology (Seminar), FS 2025

Max Felix Orson Welz

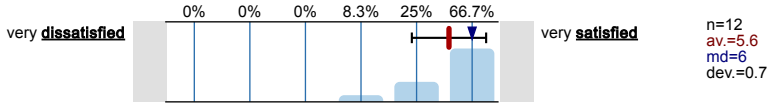
Legend

Question text

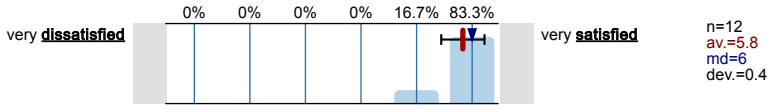


Overall Satisfaction

How **satisfied** are you with the course overall?

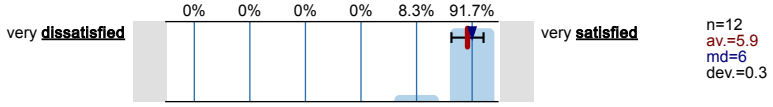


Overall, how **satisfied** are you with the **realization of the course** (e. g., course mode, portion of interactive and asynchronous elements, etc.)?

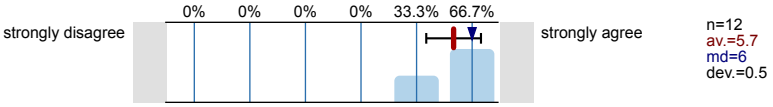


Questions on the Instructor Max Felix Orson Welz

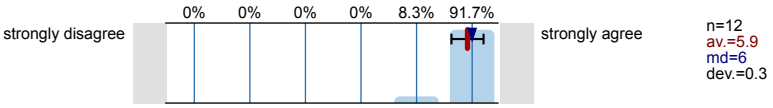
How **satisfied** are you with the instructor **overall**?



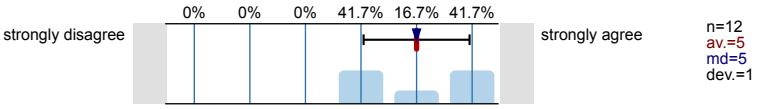
The instructor taught the content in a **clear and comprehensible** manner.



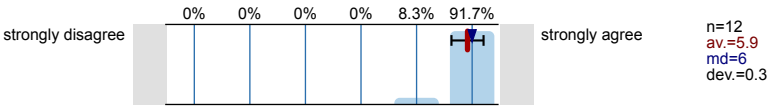
The instructor presented the lecture in a **structured way** (structure, transparency, common thread).



The instructor used various strategies to motivate the students to **actively participate** in the course units.



The instructor was **reasonably available**.



Comments regarding the **positive** aspects concerning the instructor (optional):

- 10/10 best lecturer for statistics afte Prof. Strobl. Give this guy a professorship, if he had taught us Statstics 3 last semester we would actually understand SEM.
- Good content structure and simple explanations of difficult topics.
- Max war sehr motiviert, super kompetent und hat mit grosser Begeisterung unterrichtet, was es für mich interessanter gemacht hat.
- Sehr motiviert und begeistert, deshalb hat es Spass gemacht, zuzuhören.
- Seine Art und Weise zu präsentieren ist sehr angenehm. Ich finde er hat das Beste aus den Themen herausgeholt und sie anschaulich erklärt. Auch wenn STatistik nicht immer das Lieblingsthema von jedermann ist, hat er sich wirklich Mühe gegeben uns einen guten Einblick in das Thema zu geben.
- While some topics were complex, the instructor was able to break them down really well and highlight key takeways from each topic discussed. The datasets and slides illustrated the peculiarities of dirty data really well and contributed to my understanding of the course material.

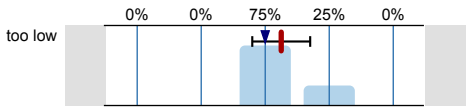
What other aspects concerning the instructor do you think need **improvement**? (Optional)

Please be aware that your comments are forwarded to the instructors without being double-checked. We kindly ask you to refrain from remarks **infringing upon the rights and honor of a person as well as inconsiderate or destructive comments.**

- Einziger Punkt zur Verbesserung ist für mich, etwas langsamer zu sprechen. Manchmal warst du so schnell, dass man dich nicht mehr richtig verstehen konnte.
- None.
- Theoretical input at the beginning was sometimes difficult to grasp, as it was quite math heavy. This is not problematic per se, but it was sometimes a bit overwhelming.

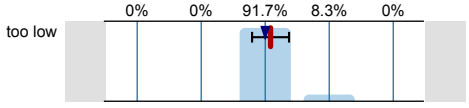
Further Opinions

How do you rate the **content-related level of difficulty** of this course? *(middle check box = "just right")*



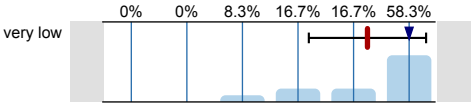
n=12
av.=3.3
md=3
dev.=0.5

How do you rate your **workload** of this course? *(middle check box = "just right")*



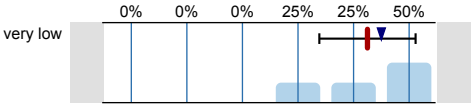
n=12
av.=3.1
md=3
dev.=0.3

How do you rate the **value of this course for your overall studies**?



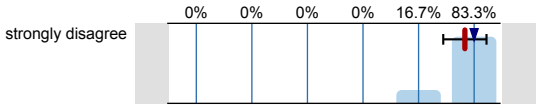
n=12
av.=5.3
md=6
dev.=1.1

How do you rate your **own acquisition of competencies** of this course?



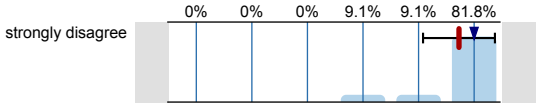
n=12
av.=5.3
md=5.5
dev.=0.9

The **learning objectives** (e.g., required expertise, social skills and self-competence) of the course were transparent.



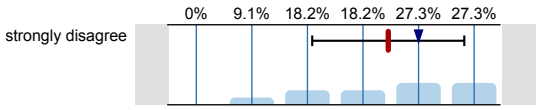
n=12
av.=5.8
md=6
dev.=0.4

The taught **competencies** will help me in my **further studies**.



n=11
av.=5.7
md=6
dev.=0.6

The taught **competencies** can help me in my **future professional activity**.



n=11
av.=4.5
md=5
dev.=1.4
ab.=1

Do you have any **comments** regarding these further opinions? (Optional)

- Ich hätte diesen Kurs etwas früher im Studium machen müssen, um richtig davon profitieren zu können.
- Thanks to the course material as well as individual support from the instructor, I profited a lot from this course. Appeciate it a lot!

Final Remarks on the Course

In the following, you can leave final remarks on the course. Please focus ond the course and avoid comments concerning the instructor/s.

Comments regarding **positive aspects** of this course (optional):

- Das Seminar war auf jeden Fall besser gestaltet als die letzte Statistikvorlesung die ich hatte. Auch die Übungen haben Spass gemacht und die Daten, die wir uns angesehen haben, waren inhaltlich spannend.
- Die Atmosphäre während der Veranstaltung war sehr angenehm, wahrscheinlich einer der besten Seminare während des Psychologiestudiums, weil unter anderem der Stoff verständlich erklärt und auf angenehmerweise vermittelt wurde, der Dozierende gut auf Fragen eingegangen war (Verständnis, weiterführende Gedanken, Administrative Fragen, etc.). Auch finde ich die Aufteilung von Theorie und Praxis während des Seminars sehr gut. Man konnte das was wir gehört haben, gleich anwenden und es nachvollziehen. Mir hat es auch gefallen dass die Aufgaben so strukturiert waren, das man zuerst den Code vom Theorieteil selbst anwendet und erst dann bei der zweiten Aufgabe es selbst bei einem anderen Beispiel anwenden muss.

- Super Aufbau, Folien sehr verständlich und vollständig, sympathische Art und Weise wie mit den Studierenden umgegangen wird. Bitte weiter so machen.
- The course was great! The instructor wrapped the course material in historical context and often used research from social sciences. Thus, I did not only learn a lot about handling dirty data and statistics in general, but also interesting topics I haven't encountered yet. This made the course really engaging and fun. I will highly recommend this course to my colleagues.

Please elaborate on aspects in **need of improvement** (e. g., ideas on the course content, learning objectives):

- Der Stoff ist zum teil sehr theoretisch (schwer greifbar).
- Die Aufgaben waren meiner Meinung nach eher z.T. einfach bzw. zu kurz. Es ist mir bewusst, dass die meisten die diesen Kurs belegen, die Masterarbeit in diesem Semester abgegeben müssen und daher nicht wirklich viel Zeit haben für anderes. Für die Personen, die aber mehr Zeit zur Verfügung haben oder auch einfach das Interesse haben mehr zu erfahren, fände ich es nicht schlecht, wenn es jeweils auch immer 1-2 schwerere / weiterführende optionale Aufgaben gegeben hätte. Persönlich mag ich den Statistik Teil in der Psychologie und habe Spass Statistik-Aufgaben zu lösen, daher fände ich es super, wenn es mehr optionale Aufgaben for fun gegeben hätte zum lösen... Ein anderer Punkt ist noch, dass ich es besser gefunden hätte, wenn wir für die letzten drei Termine auch noch weiterhin etwas theoretisches angeschaut hätten und dann den Leistungsnachweis anstatt auf Ende Mai auf Ende Juni angesetzt hätten. Die, die die Masterarbeit in diesem Semester abgegeben müssen, müssen dies i.d.R. spätestens am 1.Juni abgegeben. Gleich in der letzten Vorlesungswoche oder eine Woche danach findet die Masterprüfung statt. Wenn man nun den Leistungsnachweis auf Ende Juni verschiebt, hätte man wahrscheinlich mehr Zeit für die Masterprüfung zu lernen und die Masterarbeit zu finalisieren und nach der Prüfung bzw. Abgabe genügend Zeit sich in Ruhe noch auf den Leistungsnachweis dieser Veranstaltung zu fokussieren. Bonus: die, die MA nicht diesen Semester abgeben müssen und/oder die Prüfung haben, können mehr über dirty data erfahren und Aufgaben dazu lösen :)
- Grading as usual happens after this survey, so if the grading is unfair, this can't be accounted for here, and that is a general problem with these surveys that persists.
- Ich persönlich fände ein vertiefterer Einblick auch in die Mathematik hinter den Methoden interessant. Aber ich kann mir auch gut vorstellen, dass uns dafür die Grundlagen fehlen.

Personal Information/Study Information

I am attending this course in the following semester of my academic level:

n=12

3rd semester Master's program	2
4th semester Master's program	9
5th semester Master's program	1

My major is taught at the following faculty:

n=12

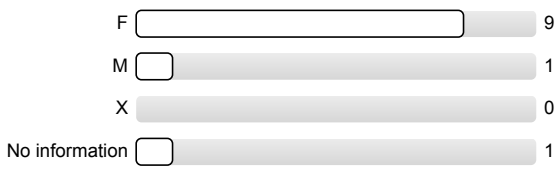
Faculty of Theology and the Study of Religion (TRF)	0
Faculty of Law (RWF)	0
Faculty of Business, Economics and Informatics (WWF)	0
Faculty of Medicine (MeF)	0
Vetsuisse Faculty (VSF)	0
Faculty of Arts and Social Sciences (PhF)	12
Faculty of Sciences (MNF)	0
I'm a student at PHZH	0
I'm attending the Teaching Diploma for Upper Secondary Education study program	0
Other	0

I am attending this course within the following study program (subject) and applying the ECTS credits to my:

n=12

Major: 180 ECTS	0
Major: 150 ECTS	0
Major: 120 ECTS	12
Major: 105 ECTS	0
Major: 90 ECTS	0
Major/Minor: 45/75 ECTS	0
Minor: 60 ECTS	0
Minor: 30 ECTS	0
Minor: 15 ECTS	0
I'm a student at PHZH	0
Other	0

Gender



Profile

Subunit: LVE - Lehrveranstaltungsevaluation/Course Evaluation
Name of the instructor: Max Felix Orson Welz
Name of the course: (Name of the survey) Dirty Data: Outliers and Missing Values in Statistics and Psychology

Values used in the profile line: Mean

Overall Satisfaction

How **satisfied** are you with the course overall?	very **dissatisfied**	very **satisfied**	n=12	av.=5.6	md=6.0	dev.=0.7
Overall, how **satisfied** are you with the **realization of the course** (e. g., course mode, portion of interactive and asynchronous elements, etc.)?	very **dissatisfied**	very **satisfied**	n=12	av.=5.8	md=6.0	dev.=0.4

Questions on the Instructor Max Felix Orson Welz

How **satisfied** are you with the instructor **overall**?	very **dissatisfied**	very **satisfied**	n=12	av.=5.9	md=6.0	dev.=0.3
The instructor taught the content in a **clear and comprehensible** manner.	strongly disagree	strongly agree	n=12	av.=5.7	md=6.0	dev.=0.5
The instructor presented the lecture in a **structured way** (structure, transparency, common thread).	strongly disagree	strongly agree	n=12	av.=5.9	md=6.0	dev.=0.3
The instructor used various strategies to motivate the students to **actively participate** in the course units.	strongly disagree	strongly agree	n=12	av.=5.0	md=5.0	dev.=1.0
The instructor was **reasonably available**.	strongly disagree	strongly agree	n=12	av.=5.9	md=6.0	dev.=0.3

Further Opinions

How do you rate the **content-related level of difficulty** of this course? *(middle check box = "just right")*	too low	too high	n=12	av.=3.3	md=3.0	dev.=0.5
How do you rate your **workload** of this course? *(middle check box = "just right")*	too low	too high	n=12	av.=3.1	md=3.0	dev.=0.3
How do you rate the **value of this course for your overall studies**?	very low	very high	n=12	av.=5.3	md=6.0	dev.=1.1
How do you rate your **own acquisition of competencies** of this course?	very low	very high	n=12	av.=5.3	md=5.5	dev.=0.9
The **learning objectives** (e.g., required expertise, social skills and self-competence) of the course were transparent.	strongly disagree	strongly agree	n=12	av.=5.8	md=6.0	dev.=0.4
The taught **competencies** will help me in my **further studies**.	strongly disagree	strongly agree	n=11	av.=5.7	md=6.0	dev.=0.6
The taught **competencies** can help me in my **future professional activity**.	strongly disagree	strongly agree	n=11	av.=4.5	md=5.0	dev.=1.4

University of Zurich[UZH]

Educational Development

University of Zurich
Educational Development
Quality Management Education & Teaching
Hirschengraben 84
CH-8001 Zurich
www.le.uzh.ch/en

Private and Confidential

University of Zurich
VP Edu / Individuelle Lehrveranstaltungsevaluation
To the instructors of the course:
"Dirty Data: Outliers and Missing Values in Statistics and Psychology"

Zurich in spring 2026

Evaluations of Teaching: Evaluation Report, Spring Semester 2026

Dear instructor(s)

Hereby you will find your evaluation report on results of your course:

„Dirty Data: Outliers and Missing Values in Statistics and Psychology“

The report contains frequencies of particular satisfaction aspects. Thus, features which need improvement from the students' perspective become recognizable. The satisfaction items with a scale show how satisfied the students are with the course overall. Furthermore, there are student comments included, if available.

Please feel free to contact us if you have any questions.

Sincerely
Quality Management Education & Teaching

These results are sent to the following instructor(s):

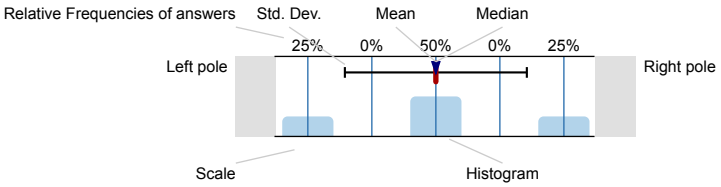
Max Felix Orson Welz

Dirty Data: Outliers and Missing Values in Statistics and Psychology (Seminar), FS 2026

Max Felix Orson Welz

Legend

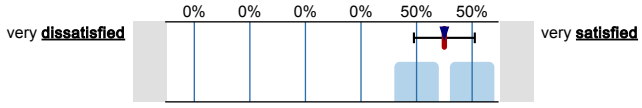
Question text



n=No. of responses
av.=Mean
md=Median
dev.=Std. Dev.
ab.=Abstention

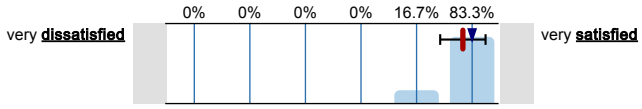
Overall Satisfaction

How **satisfied** are you with the course overall?



n=6
av.=5.5
md=5.5
dev.=0.5

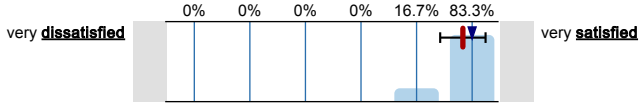
Overall, how **satisfied** are you with the **realization of the course** (e. g., course mode, portion of interactive and asynchronous elements, etc.)?



n=6
av.=5.8
md=6
dev.=0.4

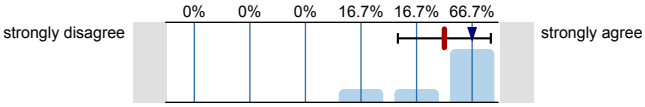
Questions on the Instructor Max Felix Orson Welz

How **satisfied** are you with the instructor **overall**?



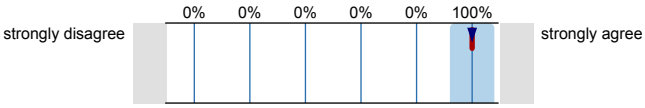
n=6
av.=5.8
md=6
dev.=0.4

The instructor taught the content in a **clear and comprehensible** manner.



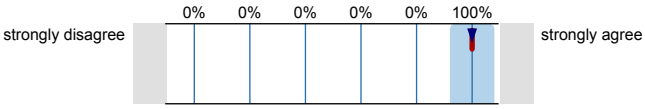
n=6
av.=5.5
md=6
dev.=0.8

The instructor presented the lecture in a **structured way** (structure, transparency, common thread).



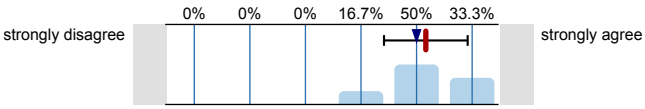
n=6
av.=6
md=6
dev.=0

The instructor displays **convincing professional expertise**.



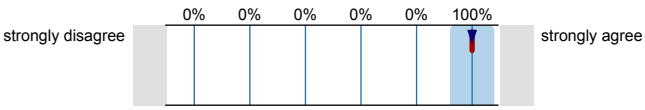
n=6
av.=6
md=6
dev.=0

The instructor used various strategies to motivate the students to **actively participate** in the course units.



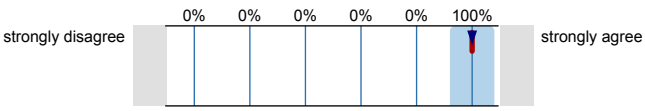
n=6
av.=5.2
md=5
dev.=0.8

The instructor responded to **questions and concerns** of the participants.



n=6
av.=6
md=6
dev.=0

The instructor was **reasonably available**.



n=5
av.=6
md=6
dev.=0
ab.=1

Comments regarding the **positive** aspects concerning the instructor (optional):

- great exercise and lecture structure
- Ich habe den Dozierenden als sehr kompetent, hilfsbereit und transparent wahrgenommen. Die Inhalte wurden verständlich erklärt.
- Lectures all followed the same structure which helped remaining engaged. The instructor used practical (and experimentative :-)) examples to highlight the content of the course and encouraged students to participate. I apreciated the fact that we could use the second half of the sessions to deepen the theoretical knowledge by completing exercises. During these, the instructor was very much present to support

students with potential questions.

- Sehr engagiert und freundlich!
  Hat Fragen immer sofort beantwortet und war sehr proaktiv!

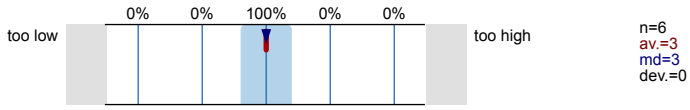
What other aspects concerning the instructor do you think need **improvement**? (Optional)

Please be aware that your comments are forwarded to the instructors without being double-checked. We kindly ask you to refrain from remarks **infringing upon the rights and honor of a person as well as inconsiderate or destructive comments.**

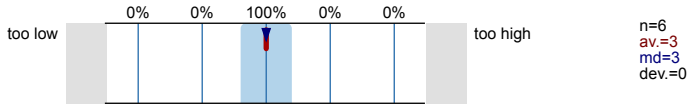
- -
- I am not sure why us students sometimes were not very responsive to questions / invitations by the instructor to engage ourselves in discussions. I think there is sometimes a bit of a barrier to do so in statistic seminars; at least I often feel I lack knowledge to answer questions. It might be helpful to try to get everyone engaged early on in the seminar (maybe with a quiz, discussion, or similar) to break the ice a bit and make students feel a bit more comfortable with speaking up.
- Maybe to go less deep into the math behind certain aspects when not really relevant. Have it on the slides but do not bother too much in the lecture.

Further Opinions

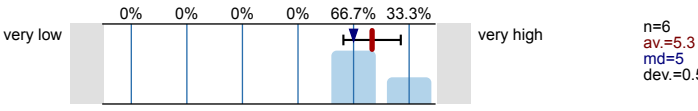
How do you rate the **content-related level of difficulty** of this course? *(middle check box = "just right")*



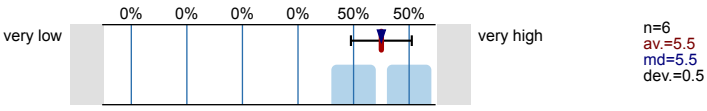
How do you rate your **workload** of this course? *(middle check box = "just right")*



How do you rate your **own acquisition of competencies** of this course?



How do you rate the **value of this course for your overall studies** ?



Do you have any **comments** regarding these further opinions? (Optional)

- Das Seminar war zusätzlich zum Hauptinhalt auch sehr hilfreich, R-Kenntnisse nochmals aufzufrischen. Dass der R-Code immer vollständig zur Verfügung gestellt wurde, hat sehr geholfen.
- Ich fand es super, dass wir das Gelernte an echten Datensätzen anwenden durften und der Dozent immer Beispiele aus echten Studien gezeigt hat. Das hat die Relevanz des Themas gut aufgezeigt.
- Knowledge can be applied to master thesis. Instructor himself made this connection throughout the course and also answered questions specific to our thesis - which was highly appreciated and helpful

Final Remarks on the Course

Comments regarding **positive aspects** of this course (optional):

- Directly applicable to master thesis. Logical building of topics.
- great course alongside your thesis, does not take too much time but you still gain relevant skills
- Ich finde, dass es eine gute Mischung aus Theorie und Anwendung war. Durch die Verwendung echter Daten wusste man, wofür man es braucht.
  Auch fand ich super, dass wir jede Woche klare Lösungen zu den Übungen erhalten haben.

Please elaborate on aspects in **need of improvement** (e. g., ideas on the course content, learning objectives):

- Insgesamt eine sehr gute Veranstaltung. Hohe Kompetenz, Transparenz, klare Folien und Anforderungen.

Personal Information/Study Information

Reason for attending this course (check all answers that apply):

n=6

Required course	0
ECTS credits	6
Interest in topic	4
Instructor	1
Other	0

I am attending this course in the following semester of my academic level:

n=6

4th semester Master's program	5
5th semester Master's program	1

My major is taught at the following faculty:

n=6

Faculty of Theology and the Study of Religion (TRF)	0
Faculty of Law (RWF)	0
Faculty of Business, Economics and Informatics (WWF)	0
Faculty of Medicine (MeF)	0
Vetsuisse Faculty (VSF)	0
Faculty of Arts and Social Sciences (PhF)	6
Faculty of Sciences (MNF)	0
I'm a student at PHZH	0
I'm attending the Teaching Diploma for Upper Secondary Education study program	0
Other	0

I am attending this course within the following study program (subject) and applying the ECTS credits to my:

n=6

Major: 180 ECTS	0
Major: 150 ECTS	0
Major: 120 ECTS	6
Major: 105 ECTS	0
Major: 90 ECTS	0
Major/Minor: 45/75 ECTS	0
Minor: 60 ECTS	0
Minor: 30 ECTS	0
Minor: 15 ECTS	0
I'm a student at PHZH	0
Other	0

Gender

n=6

F	4
M	1
Non-binary	1
No information	0

Profile

Subunit: VP Edu / Individuelle Lehrveranstaltungsevaluation
Name of the instructor: Max Felix Orson Welz
Name of the course: (Name of the survey) Dirty Data: Outliers and Missing Values in Statistics and Psychology

Values used in the profile line: Mean

Overall Satisfaction

How **satisfied** are you with the course overall?	very **dissatisfied**	very **satisfied**	n=6	av.=5.5	md=5.5	dev.=0.5
Overall, how **satisfied** are you with the **realization of the course** (e. g., course mode, portion of interactive and asynchronous elements, etc.)?	very **dissatisfied**	very **satisfied**	n=6	av.=5.8	md=6	dev.=0.4

Questions on the Instructor Max Felix Orson Welz

How **satisfied** are you with the instructor **overall**?	very **dissatisfied**	very **satisfied**	n=6	av.=5.8	md=6	dev.=0.4
The instructor taught the content in a **clear and comprehensible** manner.	strongly disagree	strongly agree	n=6	av.=5.5	md=6	dev.=0.8
The instructor presented the lecture in a **structured way** (structure, transparency, common thread).	strongly disagree	strongly agree	n=6	av.=6	md=6	dev.=0
The instructor displays **convincing professional expertise**.	strongly disagree	strongly agree	n=6	av.=6	md=6	dev.=0
The instructor used various strategies to motivate the students to **actively participate** in the course units.	strongly disagree	strongly agree	n=6	av.=5.2	md=5	dev.=0.8
The instructor responded to **questions and concerns** of the participants.	strongly disagree	strongly agree	n=6	av.=6	md=6	dev.=0
The instructor was **reasonably available**.	strongly disagree	strongly agree	n=5	av.=6	md=6	dev.=0

Further Opinions

How do you rate the **content-related level of difficulty** of this course? *(middle check box = "just right")*	too low	too high	n=6	av.=3	md=3	dev.=0
How do you rate your **workload** of this course? *(middle check box = "just right")*	too low	too high	n=6	av.=3	md=3	dev.=0
How do you rate your **own acquisition of competencies** of this course?	very low	very high	n=6	av.=5.3	md=5	dev.=0.5
How do you rate the **value of this course for your overall studies** ?	very low	very high	n=6	av.=5.5	md=5.5	dev.=0.5

2021

Academic Skills (econometrics) (FEB22015X)

Practical information

Course ID
FEB22015X

Credits
4 EC

Instruction language
Dutch

Themes

- Bachelor 2 - International Bachelor Econometrics and Operations Research

Registration procedure

Via SIN-Online

Course description

Content

This course teaches and trains writing, presentation and research skills.

Learning goals

- Students are able to present and report an economic, econometric or OR problem.
- Students are able to present and report the methods used for analysing and solving an economic, econometric or OR problem.
- Students are able to present and report the results of analysing and solving an economic, econometric or operations research problem.

Changes as compared to previous year
The content of this course is the same as last year. The treatment of topics has been slightly adjusted, inspired by conversations with and remarks of students, by past experiences with the course process, and from discussions with fellow lecturers.

Good to know
This course is connected to FEB22009X Introductory Seminar Case Studies Econometrics and Operations Research. It is not possible to follow Academic Skills without following the Introductory Seminar Case Studies.

Extra
This course should be taken jointly with FEB22009X Introductory Seminar Case Studies Econometrics.

Study materials

Required material

- Book
  Karreman, B., Werner, G., Van der Molen, H., Osseweijer, E. M., Ackermann, M., Schmidt, H., & Van der Wal, E. (2018). Academic Writing Skills for Economics and Business Administration, Boom uitgevers Amsterdam, (ISBN 978-90-5875-809-5).

Entrance requirements

Entrance requirements course enrolment

You must meet the following requirements

- Enrolled for one of the following degree programmes
  - BSc2 Econometrics
  - International Bachelor Econometrics and Operations Research

Instructional modes

Instructional modes

- Skills

Period

Enrolment periods

- Block BLOK4 until Block BLOK5
  Period of education
  28 februari 2022 until 25 juni 2022

Lecturers

Contactperson for the course

- dr. LE Westerink - Duijzer

Coordinator

- dr. LE Westerink - Duijzer

Lecturer

- dr. LE Westerink - Duijzer

Vaardighedendocent

- A Venes Schmidt, MSc

- MFO Welz

Assessment

Assessment type(s)

Examination

- assignments, essays, ...: 100%

Additional information

- Obligatory attendance at 70% of the sessions (active participation required).

Re-examination

- assignments, essays, ...: 100%

Additional information

- Assignments can be retaken during the course.

Assessment requirements

- Compulsory attendance at at least 70% of the tutorials /skills sessions (and active participation).

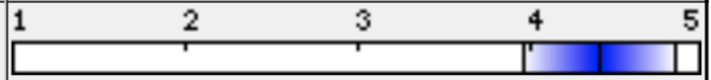
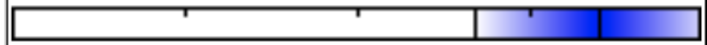
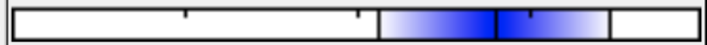
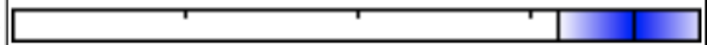
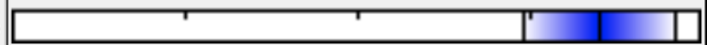
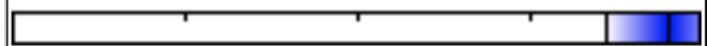
Assessment materials

- Lecture material.

Assessments (result in OSIRIS)

- Assignment(s)

  Test weight
  100

Select	Id	Question	Answers	Average	St. dev.	Graph
☐	13	The explanations given during the skills tutorials by instructor M.F.O. Welz were clear.	5	4,40	0,55	
☐	14	During the skills tutorials instructor M.F.O. Welz gave good examples to clarify the subject matter.	5	4,40	0,89	
☐	15	During the skills tutorials instructor M.F.O. Welz showed why the assignment(s) was (were) relevant.	5	3,80	0,84	
☐	16	During the skills tutorials instructor M.F.O. Welz stimulated active participation of the students.	5	4,60	0,55	
☐	17	During the skills tutorials instructor M.F.O. Welz provided sufficient assistance with the assignment(s).	5	4,40	0,55	
☐	18	During the skills tutorials instructor M.F.O. Welz provided good feedback on my work.	5	4,80	0,45	

2023

Seminar in Machine Learning (FEB23020)

Practical information

Course ID
FEB23020

Credits
12 EC

Instruction language
English

Themes

- Bachelor 3 - Econometrie en Operational Research
- Bachelor 3 - International Bachelor Econometrics and Operations Research
- Bachelor 4 - BSC2
- Major Data Science for Econometrices
- Major Econometrics
- Major Quantitative Finance
- Major Quantitative Logistics

Secretariat
ESE-Secretariat Econometrics, ET-21, 010-4081264, eb-secr@ese.eur.nl

Course description

Content

Data is known to be the oil and gold of the 21st century. However, surprisingly few professionals have enough skills to ensure the best use of this resource. With the growing availability of data, machine learning is becoming increasingly important in strategic decision-making and problem solving. Data-driven solutions can deliver benefits in virtually every industry ranging from credit scoring models and image recognition to self-driving cars and content selection on Netflix. Various machine learning techniques can uncover highly complex relationships in data which can be used for the strategic decision-making process. With the help of insights gained from structured or unstructured data, organizations hope to gain a competitive advantage and efficiency.
In this seminar, students will be introduced to the broad field of machine learning with its main concepts and principles. The students will be introduced to such central methods of machine learning as penalized regression, tree-based methods, and neural networks. Subsequently, methods for text analytics and cluster analysis will be discussed. All methods will be illustrated by concrete practical examples. Finally, the students will apply the gained knowledge to work on their own projects. After finalizing this seminar, students will acquire the necessary skills to consider data science problems from different angles, including managerial and technical.
As a result of successful studies students:

- will be able to solve real-world problems by identifying a suitable machine learning tool;

- will have the necessary skills to analyze the validity of machine learning algorithms for different types of data (binary, categorical, discrete, numerical, text, image, audio);
- will be able to select most suitable techniques by performing evaluation of learning algorithms and model selection;
- will create policy recommendations for companies by transforming results produced by machine learning algorithms.

This course uses knowledge from FEB22009(X,Q) Introductory Seminar Case Studies, FEB22002(X) Introduction to Multivariate Statistics and FEB22005(X) Econometrics 2.

Learning goals

- Obtain in-depth knowledge about econometric and OR techniques related to machine learning, and new developments in this field.
- Learn how to formulate a research question.
- Learn how to critically read and discuss academic papers.
- Learn to decide upon the appropriate econometric techniques to answer the research question.
- Learn how to apply quantitative models to data.
- Learn how to interpret results to answers to research questions.
- Apply quantitative techniques in a real-world setting.
- Learn to report about the research project in the form of a scientific report and a presentation.
- Learn to collaborate on a research project in a team with fellow students.

Changes as compared to previous year

The exploratory phase and the grading scheme have changed based on experiences last year.

Study materials

Required material

- Literature
  Articles and/or book chapters announced on Canvas.

Recommended material

- Literature
  Articles and/or book chapters announced on Canvas.

Entrance requirements

Assumed previous knowledge

- Bachelor-1 Econometrie en Operationele Research / International Bachelor Econometrics and Operational Research, and
- Bachelor-2 Econometrie en Operationele Research / International Bachelor Econometrics and Operational Research
- BSc2 students: Year 1 and Year 2, and Year 3 except Applied Microeconomics, Economics of the Welfare State, and the major IBEB.

Entrance requirements course enrolment

You must meet the following requirements

- Enrolled for one of the following degree programmes
  - BSc2 Econometrics
  - Econometrics and Operations Research
  - International Bachelor Econometrics and Operations Research

Instructional modes

Instructional modes

- Seminar

Period

Enrolment periods

- Block BLOK4M
  Period of education
  4 maart 2024 until 27 april 2024

Lecturers

Contactperson for the course

- dr. A Tetereva

Coordinator

- dr. A Tetereva
- dr. HJWG Kole

Lecturer

- dr. MA Mankowski
- dr. MH Akyuz
- H Deng
- JMJ van Druten
- M Mueller
- MFO Welz

Assessment

Assessment type(s)

Examination

	weight	type	content
Assignments/ Essays/...	100 %	set-up	• Assignment (group) • Presentation (group)

Re-examination

	weight	type	content
Assignments/ Essays/...	100 %	set-up	• Repair / Extend original assigment

Assessment requirements

Bachelor-1 Econometrie en Operationele Research / International Bachelor Econometrics and Operational Research, and, Bachelor-2 Econometrie en Operationele Research / International Bachelor Econometrics and Operational Research
BSc2 students: year 1 except Philosophy of Economics, year 2 and year 3 except Applied Microeconomics, Economics of the Welfare State, and Major IBEB

Compulsory attendance at all plenary meetings, as well as weekly appointments with supervisor
Compulsory presentation during final session

Assessment materials

Collection of journal articles and working papers, depending on the research topic.

Assessments (result in OSIRIS)

- Assignment(s)

  Test weight
  100

2022

Machine Learning (FEM31002)

Practical information

Course ID
FEM31002

Credits
4 EC

Instruction language
English

Themes

- MSc Analytics and Operations Research in Logistics
- MSc Data Science for Econometrics

Registration procedure

Via SIN-Online.

Secretariat
ESE-Secretariat Econometrics, ET-21, 010-4081264, eb-secr@ese.eur.nl

Course description

Content

While traditional statistical and econometric methods often rely on strict model assumptions, methods from statistical learning and machine learning can provide more flexibility. This flexibility is achieved by a nonparametric nature of machine learning techniques, but also through so-called hyperparameters, which have to be tuned via validation procedures.

Main topics covered in this course:
1. Resampling and Model Evaluation
2. Regularization
3. Trees and Ensembles
4. Neural Networks
5. Support Vector Machines
6. Clustering

The focus of this course is to develop a theoretical understanding of the discussed methodology. In addition, possible applications in different fields of economics, finance, and operations research will be illustrated by discussing recent journal articles and working papers. Students will gain deeper insights through mathematical exercises, as well as practical experience with applying the techniques in R or Python through hands-on exercises.

Learning goals

- Understanding the workings of fundamental supervised and unsupervised machine learning techniques
- Gaining deeper insights into the statistical theory behind the methods
- Applying the methods in R or Python

- Evaluating the performance of the methods in a systematic way

Changes as compared to previous year

- Format: in-person lectures (possibly recorded) with additional short videos instead of full video lectures
- Content: while the six major topics remain unchanged, there are some changes with respect to which techniques are discussed for each topic

Extra

Two intermediate reports and a final exam.

Study materials

Required material

- Book
  - "The Elements of Statistical Learning: Data Mining, Inference, and Prediction." T. Hastie, R. Tibshirani, J. Friedman, Second Edition, Springer, 2009.
- Canvas
  - Selected papers (see Canvas)

Entrance requirements

Assumed previous knowledge

Only accessible for Master's students with a valid Admission Statement.

Entrance requirements course enrolment

You must meet the following requirements

- Enrolled for one of the following degree programmes
  - Econometrics and Management Science
- In Block BLOK1 you may not be part of one of the following target groups
  - BLOKKADE

Instructional modes

Instructional modes

- Lecture
- Question hour

Period

Enrolment periods

- Block BLOK1
  Period of education
  29 augustus 2022 until 24 oktober 2022

Lecturers

Contactperson for the course

- dr. A Alfons

Coordinator

- dr. A Alfons

Lecturer

- dr. A Alfons
- dr. MH Akyuz
- MFO Welz
- prof.dr. DJC van Dijk

Assessment

Assessment type(s)

Examination

	weight	type	content
Assignments/ Essays/...	30 %	set-up	• Assignment (group)
Written exam	70 %	number	1
		type	Written exam
		set-up	• The exam contains open questions

Re-examination

	weight	type	content
Assignments/ Essays/...	30 %	set-up	• Use original grade again
Written exam	70 %	number	1
		type	Written exam
		set-up	• The re-exam contains open questions

Assessment materials

- Selected chapters of: T. Hastie, R. Tibshirani, and J. Friedman. The Elements of Statistical Learning: Data Mining, Inference, and Prediction. Second Edition, Springer, 2009. https://hastie.su.domains/ElemStatLearn/

- Selected papers
- All other course materials (except when explicitly excluded)
- See Canvas for more details

Assessments (result in OSIRIS)

- written exam

  Test weight
  100

Test enrolment

written exam

- Block BLOK1
  Enrolment period
  9 September 2022 until 10 October 2022 23:59

  Late enrolment
  11 October 2022 until 17 October 2022 23:59

  Unenrolment
  29 August 2022 until 13 October 2022 23:59

  Test date
  17 October 2022
- Block BLOK-HER
  Enrolment period
  1 June 2023 until 2 July 2023 23:59

  Late enrolment
  3 July 2023 until 7 July 2023 23:59

  Unenrolment
  17 May 2023 until 5 July 2023 23:59

  Test date
  7 July 2023

2021

Seminar in Econometrics (FEB23012)

Practical information

Course ID
FEB23012

Credits
12 EC

Instruction language
English

Themes

- Bachelor 3 - Econometrie en Operational Research
- Bachelor 3 - International Bachelor Econometrics and Operations Research
- Bachelor 4 - BSC2
- Major Econometrics

Registration procedure

Via centrale inschrijving

Secretariat
ESE-Secretariat Econometrics, ET-21, 010-4081264, eb-secr@ese.eur.nl

Course description

Content

During this seminar groups of 3 or 4 students will try to answer a complex practical research question using econometric knowledge obtained during earlier courses. They have to report the results of the research in a scientific report and in a presentation.

Learning goals

- The student learns how to formulate a proper research question with testable hypotheses.
- The student learns to decide upon the correct data and appropriate econometric model to answer the research question.
- The student learns how to do a research project in a team of four students.
- The student learns to apply his econometric knowledge to do proper econometric inference.
- The student learns to use estimation output resulting from an econometric analysis to answer the research question and to report his research in a scienfic report and a presentation.

Study materials

Required material

- Article
  Collection of articles and working papers, depending on the research project. References will be provided by the project supervisor.

Entrance requirements

Assumed previous knowledge

- Bachelor-1 Econometrie en Operationele Research / International Bachelor Econometrics and Operational Research, and,
- Bachelor-2 Econometrie en Operationele Research / International Bachelor Econometrics and Operational Research
- BSc2 students: year 1 except Philosophy of Economics, year 2 and year 3 except International Economics, Economics of the Welfare State, Game Theory and its Applications, and Major IBEB

Entrance requirements course enrolment

You must meet the following requirements

- Enrolled for one of the following degree programmes
  - BSc2 Econometrics
  - Econometrics and Operations Research
  - International Bachelor Econometrics and Operations Research

Instructional modes

Instructional modes

- Seminar

Period

Enrolment periods

- Block BLOK4
  Period of education
  28 februari 2022 until 24 april 2022

Lecturers

Contactperson for the course

- prof.dr. DJC van Dijk

Coordinator

- prof.dr. DJC van Dijk

Lecturer

- dr. EP O'Neill, PhD
- MFO Welz

Assessment

Assessment type(s)

Examination

- assignments, essays, ...: 100%

Additional information

- Compulsory attendance at all plenary meetings, as well as weekly appointments with supervisor;
- Compulsory presentation during final session.

Re-examination

- There is no re-examination

Assessment requirements

- Mandatory attendance at all sessions including the weekly meetings with the supervisor of your project

Assessment materials

- Selection of papers.

Assessments (result in OSIRIS)

- Assignment(s)

  Test weight
  100

2019

Introduction to Multivariate Statistics (FEB22003X)

Practical information

Course ID
FEB22003X

Credits
4 EC

Instruction language
English

Themes

- Bachelor 2 - BSC2
- Bachelor 2 - International Bachelor Econometrics and Operations Research
- Premaster Econometrics and Management Science (EN)

Registration procedure
Via SIN-Online

Secretariat
ESE-Secretariat Econometrics, ET-21, 010-4081264, eb-secr@ese.eur.nl

Course description

Content
The course is primarily theoretical. In this course we use a lot of linear algebra and see its connection with statistics. After the univariate statistics studied in the previous courses, we move to multivariate statistical methods. We start with the geometry of multivariate samples. Then we generalize the univariate methods to multivariate setting. And finally, we study several techniques, which are useful in practice.

Learning goals

- Students learn to use some important non-parametric tests, such as the sign test and the Mann-Whitney tests.
- Students get introduced to the multivariate normal distribution.
- Students learn to apply some mulitvariate techiques that involve the multivariate normal distribution.
- Students learn the basis of principal components and factor analysis.

Study materials

Required material

- Book
  Johnson and Wichern; Applied Multivariate Statistical Analysis, 6th edition, 2007. ISBN: 0-13-514350-0

Entrance requirements

Entrance requirements course enrolment

You must meet the following requirements

- Enrolled for one of the following degree programmes
  - BSc2 Econometrics
  - International Bachelor Econometrics and Operations Research
  - International Premaster Econometrics and Management Science

Instructional modes

Instructional modes

- Exercise Lecture
- Lecture
- Tutorial

Period

Enrolment periods

- Block BLOK1
  Period of education
  1 september 2019 until 27 oktober 2019

Lecturers

Contactperson for the course

- dr. M Zhelonkin

Coordinator

- dr. M Zhelonkin

Lecturer

- dr. M Zhelonkin

Practicumdocent

- E Boon
- JJ Verwoerd
- LA Van Duuren, MSc
- M Cheng
- MFO Welz

- N Meekma

Assessment

Assessment type(s)

- Assignments, essays... (10%)
- Written (re-)examination with essay questions (90%)

Assessment materials

Chapters from Johnson and Wichern.

Assessments (result in OSIRIS)

- Written examination

  Test weight

  100

Test enrolment

Written examination

- Block BLOK1

  Enrolment period

  16 September 2019 until 15 October 2019 23:59

  Unenrolment

  3 September 2019 until 20 October 2019 23:59

  Test date

  22 October 2019

- Block BLOK-HER

  Enrolment period

  12 June 2020 until 13 July 2020 23:59

  Unenrolment

  29 May 2020 until 16 July 2020 23:59

  Test date

  20 July 2020

- Block BLOK-EXT

  Enrolment period

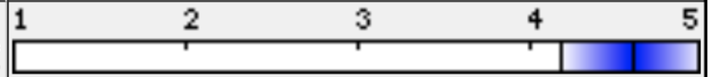
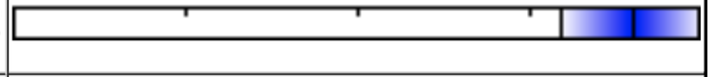
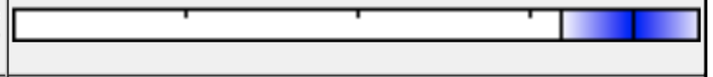
  10 July 2020 until 10 August 2020 23:59

  Unenrolment

  29 June 2020 until 13 August 2020 23:59

  Test date

  17 August 2020

Select	Id	Question	Answers	Average	St. dev.	Graph
☐	23	During the tutorials instructor M.F.O. Welz motivated me to study the subject.	6	4,67	0,52	
☐	24	During the tutorials instructor M.F.O. Welz stimulated active participation of the students.	6	4,67	0,52	
☐	25	During the tutorials instructor M.F.O. Welz provided sufficient assistance.	6	4,67	0,52	
☐	26	Instructor M.F.O. Welz has a good command of English during the tutorials.	6	4,67	0,52	